PART -A

01.Write a R Program to implement Operations of set (Union,Intersection,Difference,Subset)

```r
#Define sets
 A<-c(1,2,3,4,5)
 B<-c(4,5,6,7,8)

#Union of A and B
  union_AB<-union(A,B)
  cat("Union of A and B:",union_AB,"\n")


#Intersection of A and B
  intersection_AB<-intersect(A,B)
  cat("Intersection of A and B:",intersection_AB,"\n")

#Difference of A and B
  difference_AB<-setdiff(A,B)
  cat("Difference of A and B (A-B):",difference_AB,"\n")


#Check if A is a subset of B
  is_subset<-all(A %in% B)
  cat("Is A a subset of B?",is_subset,"\n")

  OUTPUT:
    Union of A and B: 1 2 3 4 5 6 7 8
    Intersection of A and B: 4 5
    Difference of A and B (A-B): 1 2 3
    Is A a subset of B? FALSE
```

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02.Write a R Program to Implement Inverse Functction

```
#Define a function and its inverse
   inverse_function<-function(x){
     return(1/x)
   }

#Test the inverse_function
x<-5
inverse_x<-inverse_function(x)
cat("Inverse of",x,"is",inverse_x,"\n")
```

OUTPUT:

```
  Inverse of 5 is 0.2
```

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03.Write a R Program to Implement One-to-One Function

```r
#Define function
  one_to_one_function<-function(x){
    return(2*x+3)
  }

#Test the one-to-one function
  x_values<-c(1,2,3,4)
 results<-one_to_one_function(x_values)
cat("Function values for",x_values,"are",results,"\n")
```

OUTPUT:

```
Function values for 1 2 3 4 are 5 7 9 11
```

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04.Write a R Program to implement Cartesian Product of Two set

```r
#Define sets
  A<-c(1,2)
  B<-c('a','b')

#Cartesian product
  cartesian_product<-expand.grid(A,B)
  cat("Cartesian Product of A and B:\n")
  print(cartesian_product)
```

OUTPUT:

```
    Cartesian Product of A and B:
    Var1 Var2
1    1    a
2    2    a
3    1    b
4    2    b
```

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05.Write a R Program to Check Whether the Given Relation is Reflexive

```r
#Define relation
  relation<-matrix(c(1,2,3,2,1,3,3,2,1),nrow=3,byrow=TRUE)

#Check reflexivity
reflexive<-all(diag(relation)==c(1,1,1))
cat("Is the relation reflexive?",reflexive,"\n")
```

OUTPUT:

```
  Is the relation reflexive? TRUE
```

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06. Write a R Program to check whether the Relation is Transitive

```
 #Define Relation
   relation<-matrix(c(1,0,1,0,1,0,0,0,1),nrow=3,byrow=TRUE)

#Check Transitivity
  transitive<-all((relation%*%relation)<=relation)
  cat("Is the relation transitive?",transitive,"\n")
```

OUTPUT:

```
  Is the relation transitive? FALSE
```

07. Write a R program to Check whether the given relation is symmetric

```
#Define relation
   relation<-matrix(c(1,1,0,1,1,0,0,0,1),nrow=3,byrow=TRUE)

#Check Symmetry
   symmetric<-all(relation==t(relation))
   cat("Is the relation symmetric?",symmetric,"\n")
```

OUTPUT:

```
    Is the relation symmetric? TRUE
```

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08. Write a R program to implement Logic Gates(NOT,AND,OR,XOR)

```
x<-TRUE
y<-FALSE

cat("NOT(x):",!x,"\n")
cat("AND(x,y):",x&y,"\n")
cat("OR(x,y):",x|y,"\n")
cat("XOR(x,y):",x!=y,"\n")
```

OUTPUT:

```
NOT(x): FALSE
AND(x,y): FALSE
OR(x,y): TRUE
XOR(x,y): TRUE
```

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PART-B

01.Write a R program to Calculate Central Tendency [Mean,Median,Mode]

```
data<-c(1,2,2,3,4,5,5,5,6,7,8,9,10)
uniqv<-unique(data)
freq<-tabulate(match(data,uniqv))
mode_value<-uniqv[which.max(freq)]
cat("Mean:", mean(data),"\n")
cat("Median:", median(data),"\n")
cat("Mode:", mode_value,"\n")
```

OUTPUT:

```
Mean: 5.153846
Median: 5
Mode: 5
```

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O2.Write a R Program to Calculate Simple Linear Algebra Operations

```
A<-matrix(c(1,2,3,4),nrow=2,byrow=TRUE)
B<-matrix(c(9,8,7,6),nrow=2,byrow=TRUE)

#Addition of two matrix (A + B)
  print(A+B)
```

```
  OUTPUT: [1] [2]
     [1,] 10 10
     [2,] 10 10
```

```
 #Subtraction of two matrix (A - B)
  print(A-B)
```

```
   OUTPUT:   [1] [2]
        [1,] -8 -6
        [2,] -4 -2
```

```
 #Multiplication of matrix (A X B)
   print(A%*%B)
```

```
  OUTPUT:   [1] [2]
       [1,] 23 20
       [2,] 55 48
```

```
 #Transposition of matrix A:
   print(t(A))
```

```
  OUTPUT:   [,1] [,2]
      [1,]  1    3
      [2,]  2    4
```

```
#Determinate of Matrix A:
  print(det(A))
```

```
   OUTPUT: [1] -2
```

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03.Write a R Program to Calculate Cumulative sum,product,minima,maxima

```
 data<-c(1,2,3,4,5)
#Cumulative sum
cat("Cumulative Sum:",cumsum(data),"\n")
```

OUTPUT: Cumulative Sum: 1 3 6 10 15

```
#Cumulative Product
cat("Cumulative Product:",cumprod(data),"\n")
```

OUTPUT: Cumulative Product: 1 2 6 24 120

```
#Minima and Maxima
cat("Minimum:",min(data),"\n")
cat("Maximum:",max(data),"\n")
```

OUTPUT: Minimum:1
Maximum:5

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04.Write a R Program to Calculate Arithmetic Mean for Ungrouped and Grouped data

```
#Ungrouped Data
ungrouped_data<-c(2,4,6,8,10)
mean_ungrouped<-mean(ungrouped_data)
cat("Arithmetic Mean for Ungrouped Data:",mean_ungrouped,"\n")
```

OUTPUT:

```
  Arithmetic Mean for Ungrouped Data: 6
```

```
#Grouped Data
values<-c(2,4,6,8,10)
frequencies<-c(1,2,3,2,1)
mean_grouped<-sum(values*frequencies)/sum(frequencies)
cat("Arithmetic Mean for Grouped Data:",mean_grouped,"\n")
```

OUTPUT:

```
  Arithmetic Mean for Grouped Data: 6
```

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05.Write a R Program to Calculate Coefficient of Variance for Discrete and Continuous series

```
#Discrete series
discrete_data<-c(5,6,7,8,9)
sd_discrete<-sd(discrete_data)
mean_discrete<-mean(discrete_data)
cv_discrete<-(sd_discrete/mean_discrete)*100
cat("Discrete Series Coefficient of Variation(%):",cv_discrete,"\n")
```

OUTPUT:

```
   Discrete Series Coefficient of Variation(%): 22.5877
```

```
#Continuous series
continuous_data<-c(10.5,12.3,11.4,10.7,11.9)
sd_continuous<-sd(continuous_data)
mean_continuous<-mean(continuous_data)
cv_continuous<-(sd_continuous/mean_continuous)*100
cat("Continuous Series Coefficient of Variation(%):",cv_continuous,"\n")
```

OUTPUT:

```
  Continuous Series Coefficient of Variation(%): 6.750102
```

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